There are many microbiological processes that occur in the presence of air (aerobically) yielding incomplete oxidation products. Examples: i) the formation of acetic acid (vinegar) from alcohol by vinegar bacteria ii) citric acid from sugar by certain molds such as Aspergillus niger. These microbial processes are often referred to as fermentations, although they do not decompose in the absence of air.

Infobits

The German Eduard Buchner, winner of the 1907 Nobel Prize in chemistry, determined that fermentation was actually caused by a yeast secretion that he termed zymase. The experiment for which Buchner won the Nobel Prize consisted of producing a cell–free extract of yeast cells and showing that this "press juice" could ferment sugar. This finding dealt yet another blow to vitalism by demonstrating for the first time that fermentation could occur outside living cells.

6.2 Screening of Industrially Important Microorganism

The next step after isolation of microorganisms is the selection or screening. For the successful fermentation process, selection of microorganisms is the prime important step.

The microbes used in the industrial microbiology should have following characters.

1. The strain should be a high–yielding strain.
2. The strain should have stable biochemical and genetical characteristics.
3. It should not produce undesirable substances.
4. It should be easily cultivated on large scale.
5. The strains should be able to protect themselves from contamination.

The strain should be in pure culture, free from other microorganisms including Bacteriophages. These characters are screened for the production of desirable products from microorganisms.

Isolation of industrially important microorganisms

Success of fermentation depends upon the isolation of microorganism. The microorganisms are isolated from their natural habitats like soil, lakes, river mud or even in unusual habitats or environments such as extreme cold, high altitude, deserts, and deep sea and petroleum fields and are tested directly for the product formation and isolated or it can be genetically modified. Different types of microorganisms are isolated by different methods. Different microbes with desired activity are isolated using various culture techniques. Screening includes primary screening and secondary screening.

Primary screening: The elementary steps that are performed to select the desired organisms and eliminate the undesirable organisms are termed as primary screening. Methods such as crowded plate technique, auxanography and enrichment culture technique are some of the techniques used in primary screening are used in primary screening. For screening of antibiotic producing organisms crowded plate technique is described here,

Crowded plate technique

1. It is used for the screening of antibiotic producing organisms.
2. Soil is serially diluted

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